

B1 Flows, Fluctuations and Complexity — Model Solutions, 2017

Question 1 — Surface gravity–capillary waves

Problem. Derive Laplace’s equation for the velocity potential of an incompressible irrotational flow. For deep-water linear surface waves with $\eta = A \sin(kx - \omega t)$ find ϕ , the dispersion relation, the particle paths, and (with surface tension) $\omega^2 = c_1 k + c_2 k^3$.

Laplace’s equation

Irrotational $\nabla \times \mathbf{u} = 0$ implies potential exists:

$$\mathbf{u} = \nabla \phi.$$

Take divergence:

$$\nabla \cdot \mathbf{u} = \nabla \cdot (\nabla \phi) = \nabla^2 \phi.$$

Apply incompressibility $\nabla \cdot \mathbf{u} = 0$:

$\nabla^2 \phi = 0.$

Boundary conditions

Kinematic ($\partial_t \eta = \partial_z \phi$ at $z = 0$): a fluid particle on the surface stays on the surface, so the vertical fluid velocity equals the surface vertical velocity (linearised: evaluated at $z = 0$).

Dynamic ($\partial_t \phi = -g\eta$ at $z = 0$): linearised Bernoulli at the free surface with $p = p_{\text{atm}}$ gives $\partial_t \phi + g\eta + p_{\text{atm}}/\rho = \text{const}$; absorb the constant into ϕ .

Depth condition: fluid at rest as $z \rightarrow -\infty$:

$$\nabla \phi \rightarrow 0, \quad \frac{\partial \phi}{\partial z} \rightarrow 0 \quad (z \rightarrow -\infty).$$

Velocity potential and dispersion relation

Try separable form $\pi/2$ out of phase with η :

$$\phi(x, z, t) = f(z) \cos(kx - \omega t).$$

Compute $\partial_x^2 \phi$:

$$\partial_x^2 \phi = -k^2 f(z) \cos(kx - \omega t).$$

Compute $\partial_z^2 \phi$:

$$\partial_z^2 \phi = f''(z) \cos(kx - \omega t).$$

Sum and apply $\nabla^2 \phi = 0$:

$$(f''(z) - k^2 f(z)) \cos(kx - \omega t) = 0.$$

Reduce to ODE:

$$f''(z) - k^2 f(z) = 0.$$

General solution:

$$f(z) = Ce^{kz} + De^{-kz}.$$

Apply $z \rightarrow -\infty$ decay (with $k > 0$, $e^{-kz} \rightarrow \infty$):

$$D = 0.$$

Hence:

$$\phi = Ce^{kz} \cos(kx - \omega t).$$

Compute $\partial_t \eta$ and $\partial_z \phi|_{z=0}$:

$$\partial_t \eta = -A\omega \cos(kx - \omega t), \quad \partial_z \phi|_0 = Ck \cos(kx - \omega t).$$

Equate (kinematic BC):

$$-A\omega \cos(kx - \omega t) = Ck \cos(kx - \omega t).$$

Solve for C :

$$C = -\frac{A\omega}{k}.$$

Velocity potential:

$$\phi(x, z, t) = -\frac{A\omega}{k} e^{kz} \cos(kx - \omega t).$$

 (1)

Compute $\partial_t \phi$ at $z = 0$:

$$\partial_t \phi|_0 = -\frac{A\omega^2}{k} \sin(kx - \omega t).$$

Apply dynamic BC $\partial_t \phi = -g\eta$:

$$-\frac{A\omega^2}{k} \sin(kx - \omega t) = -gA \sin(kx - \omega t).$$

Cancel $-A \sin(kx - \omega t)$:

$$\frac{\omega^2}{k} = g.$$

Multiply by k :

$$\omega^2 = gk.$$

 (2)

Particle paths

Lagrangian displacement (ξ, ζ) of a particle about (x_0, z_0) :

$$\dot{\xi} = u = \partial_x \phi, \quad \dot{\zeta} = w = \partial_z \phi.$$

Differentiate $\phi = -(A\omega/k)e^{kz} \cos(kx - \omega t)$ w.r.t. x :

$$\partial_x \phi = -\frac{A\omega}{k} e^{kz} \cdot (-k) \sin(kx - \omega t).$$

Simplify:

$$\partial_x \phi = A\omega e^{kz} \sin(kx - \omega t).$$

Differentiate w.r.t. z :

$$\partial_z \phi = -A\omega e^{kz} \cos(kx - \omega t).$$

Evaluate at the mean position (x_0, z_0) (linear theory):

$$u = A\omega e^{kz_0} \sin(kx_0 - \omega t), \quad w = -A\omega e^{kz_0} \cos(kx_0 - \omega t).$$

Integrate u in time:

$$\xi = \int A\omega e^{kz_0} \sin(kx_0 - \omega t) dt.$$

Use $\int \sin(kx_0 - \omega t) dt = \frac{1}{\omega} \cos(kx_0 - \omega t)$:

$$\xi = A e^{kz_0} \cos(kx_0 - \omega t).$$

Integrate w in time:

$$\zeta = \int -A\omega e^{kz_0} \cos(kx_0 - \omega t) dt.$$

Use $\int \cos(kx_0 - \omega t) dt = -\frac{1}{\omega} \sin(kx_0 - \omega t)$:

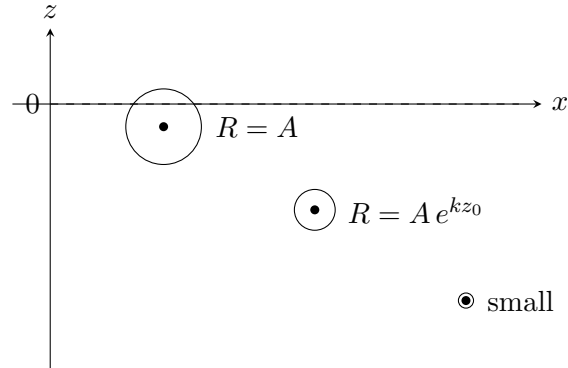
$$\zeta = A e^{kz_0} \sin(kx_0 - \omega t).$$

Square and add (apply $\sin^2 + \cos^2 = 1$):

$$\xi^2 + \zeta^2 = A^2 e^{2kz_0}.$$

Circular paths of radius $R(z_0) = A e^{kz_0}$, decaying exponentially with depth:

$R(z_0) = A e^{kz_0}.$



Surface tension: dispersion relation

Modified dynamic BC at $z = 0$:

$$\frac{\partial \phi}{\partial t} = -g\eta + \frac{\sigma}{\rho} \frac{\partial^2 \eta}{\partial x^2}.$$

Compute $\partial_x^2 \eta$:

$$\frac{\partial^2 \eta}{\partial x^2} = -Ak^2 \sin(kx - \omega t).$$

Insert $\partial_t \phi|_0 = -(A\omega^2/k) \sin(kx - \omega t)$ and η :

$$-\frac{A\omega^2}{k} \sin(kx - \omega t) = -gA \sin(kx - \omega t) - \frac{\sigma}{\rho} Ak^2 \sin(kx - \omega t).$$

Cancel $-A \sin(kx - \omega t)$:

$$\frac{\omega^2}{k} = g + \frac{\sigma k^2}{\rho}.$$

Multiply by k :

$$\boxed{\omega^2 = gk + \frac{\sigma}{\rho} k^3, \quad c_1 = g, \quad c_2 = \sigma/\rho.} \quad (3)$$

Wave speed $c = \omega/k$:

$$c^2 = \frac{\omega^2}{k^2} = \frac{g}{k} + \frac{\sigma k}{\rho}.$$

Take square root:

$$c(k) = \sqrt{\frac{g}{k} + \frac{\sigma k}{\rho}}.$$

Minimise c^2 via $d(c^2)/dk = 0$:

$$-\frac{g}{k^2} + \frac{\sigma}{\rho} = 0.$$

Solve for k_* :

$$k_* = \sqrt{g\rho/\sigma}.$$

Insert $k_* = \sqrt{g\rho/\sigma}$ into c^2 :

$$c_{\min}^2 = \frac{g}{\sqrt{g\rho/\sigma}} + \frac{\sigma}{\rho} \sqrt{g\rho/\sigma}.$$

Simplify first term:

$$\frac{g}{\sqrt{g\rho/\sigma}} = \sqrt{g\sigma/\rho}.$$

Simplify second term:

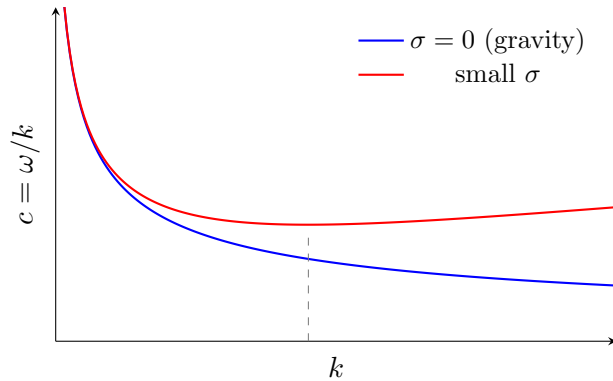
$$\frac{\sigma}{\rho} \sqrt{g\rho/\sigma} = \sqrt{g\sigma/\rho}.$$

Sum:

$$c_{\min}^2 = 2\sqrt{g\sigma/\rho}.$$

Take square root:

$$c_{\min} = (4g\sigma/\rho)^{1/4}.$$



Short waves ($k \gg k_*$, small λ) most affected: the curvature $\partial_x^2 \eta \propto k^2$ amplifies the surface-tension restoring force, while gravity g/k becomes negligible. Long waves ($k \ll k_*$) are gravity-dominated.

Question 2 — Reynolds number, cylinder with source, Bernoulli, Torricelli

Problem. Derive Re by non-dimensionalising 2-D steady NS; for $\phi = Ur \cos \theta + \mu \cos \theta / r + m \log r$ around a cylinder of radius a with outward normal velocity V , find μ, m ; for $V = 4U$ locate the upstream stagnation point; derive Bernoulli; find efflux speed and flux from a tank.

Reynolds number

Steady 2-D incompressible NS (drop gravity / absorb into pressure):

$$(\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}.$$

Introduce scales $\mathbf{u} = U \mathbf{u}^*$, $\nabla = L^{-1} \nabla^*$, $p = \rho U^2 p^*$:

$$\frac{U^2}{L} (\mathbf{u}^* \cdot \nabla^*) \mathbf{u}^* = -\frac{1}{\rho} \cdot \frac{\rho U^2}{L} \nabla^* p^* + \frac{\nu U}{L^2} \nabla^{*2} \mathbf{u}^*.$$

Cancel ρ in the pressure term:

$$\frac{U^2}{L} (\mathbf{u}^* \cdot \nabla^*) \mathbf{u}^* = -\frac{U^2}{L} \nabla^* p^* + \frac{\nu U}{L^2} \nabla^{*2} \mathbf{u}^*.$$

Divide through by U^2/L :

$$(\mathbf{u}^* \cdot \nabla^*) \mathbf{u}^* = -\nabla^* p^* + \frac{\nu}{UL} \nabla^{*2} \mathbf{u}^*.$$

Identify dimensionless coefficient $1/\text{Re}$:

$$(\mathbf{u}^* \cdot \nabla^*) \mathbf{u}^* = -\nabla^* p^* + \frac{1}{\text{Re}} \nabla^{*2} \mathbf{u}^*.$$

Hence:

$$\boxed{\text{Re} = \frac{UL}{\nu}}.$$

Behaviour: $\text{Re} \ll 1$ Stokes (creeping) flow, viscosity dominates, time-reversible streamlines. $\text{Re} \sim O(1)$ steady laminar flow with separation possible. $\text{Re} \gtrsim 40$ unsteady wake (von Kármán street). $\text{Re} \gtrsim 10^3$ – 10^5 transition to turbulence; inertia dominates, viscous effects confined to thin boundary layers.

Cylinder with source: μ, m

Compute $u_r = \partial \phi / \partial r$:

$$u_r = \frac{\partial}{\partial r} \left[Ur \cos \theta + \frac{\mu \cos \theta}{r} + m \log r \right].$$

Differentiate term-by-term:

$$u_r = U \cos \theta - \frac{\mu \cos \theta}{r^2} + \frac{m}{r}.$$

Compute $u_\theta = (1/r) \partial \phi / \partial \theta$:

$$u_\theta = \frac{1}{r} \frac{\partial}{\partial \theta} \left[Ur \cos \theta + \frac{\mu \cos \theta}{r} + m \log r \right].$$

Differentiate term-by-term:

$$u_\theta = \frac{1}{r} \left[-Ur \sin \theta - \frac{\mu \sin \theta}{r} \right] = -U \sin \theta - \frac{\mu \sin \theta}{r^2}.$$

Apply normal BC $u_r(a, \theta) = V$ for all θ :

$$U \cos \theta - \frac{\mu \cos \theta}{a^2} + \frac{m}{a} = V.$$

Match $\cos \theta$ coefficient:

$$U - \frac{\mu}{a^2} = 0.$$

Solve for μ :

$$\mu = Ua^2.$$

Match constant (θ -independent) part:

$$\frac{m}{a} = V.$$

Solve for m :

$$m = Va.$$

$$\boxed{\mu = Ua^2, \quad m = Va.}$$

Potential and velocities:

$$\begin{aligned} \phi &= U \cos \theta \left(r + \frac{a^2}{r} \right) + Va \log r, \\ u_r &= U \cos \theta \left(1 - \frac{a^2}{r^2} \right) + \frac{Va}{r}, \quad u_\theta = -U \sin \theta \left(1 + \frac{a^2}{r^2} \right). \end{aligned}$$

Stagnation point for $V = 4U$

By symmetry, the stagnation point lies on $\theta = \pi$ (upstream axis). On this axis $\sin \theta = 0$ so $u_\theta \equiv 0$; require $u_r = 0$ with $V = 4U$:

$$U \cos \pi \left(1 - \frac{a^2}{r^2} \right) + \frac{4Ua}{r} = 0.$$

Substitute $\cos \pi = -1$:

$$-U \left(1 - \frac{a^2}{r^2} \right) + \frac{4Ua}{r} = 0.$$

Cancel U :

$$-\left(1 - \frac{a^2}{r^2} \right) + \frac{4a}{r} = 0.$$

Multiply by r^2 :

$$-r^2 + a^2 + 4ar = 0.$$

Rearrange:

$$r^2 - 4ar - a^2 = 0.$$

Apply quadratic formula:

$$r = \frac{4a \pm \sqrt{16a^2 + 4a^2}}{2}.$$

Simplify the discriminant:

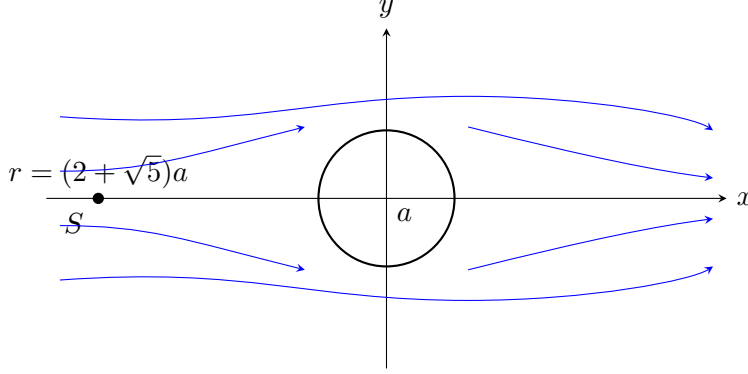
$$r = \frac{4a \pm \sqrt{20a^2}}{2} = \frac{4a \pm 2a\sqrt{5}}{2}.$$

Divide by 2:

$$r = 2a \pm a\sqrt{5} = (2 \pm \sqrt{5})a.$$

Take the root with $r > a$ (outside cylinder); $(2 - \sqrt{5})a < 0$ is rejected:

$$r_{\text{stag}} = (2 + \sqrt{5})a.$$



Bernoulli along streamlines

Steady, inviscid, incompressible Euler with gravity:

$$(\mathbf{u} \cdot \nabla)\mathbf{u} + \frac{1}{\rho}\nabla p + g\hat{\mathbf{k}} = 0.$$

Write $g\hat{\mathbf{k}} = \nabla(gz)$:

$$(\mathbf{u} \cdot \nabla)\mathbf{u} + \frac{1}{\rho}\nabla p + \nabla(gz) = 0.$$

Apply identity $(\mathbf{u} \cdot \nabla)\mathbf{u} = \nabla(\frac{1}{2}u^2) + \boldsymbol{\omega} \times \mathbf{u}$:

$$\nabla(\frac{1}{2}u^2) + \boldsymbol{\omega} \times \mathbf{u} + \frac{1}{\rho}\nabla p + \nabla(gz) = 0.$$

Multiply through by ρ (constant) and group gradients:

$$\nabla(p + \frac{1}{2}\rho u^2 + \rho gz) = -\rho \boldsymbol{\omega} \times \mathbf{u}.$$

Take dot product with $\mathbf{u}$ and use $\mathbf{u} \cdot (\boldsymbol{\omega} \times \mathbf{u}) = 0$:

$$\mathbf{u} \cdot \nabla B = 0, \quad B \equiv p + \frac{1}{2}\rho u^2 + \rho gz.$$

Therefore B is constant along streamlines:

$$p + \frac{1}{2}\rho u^2 + \rho gz = \text{const along streamlines.}$$

Tank: efflux speed and flux

Assumptions: steady, inviscid, incompressible, $A_e \ll A_0$ so surface speed ≈ 0 ; both surface and exit at p_{atm} .

Bernoulli surface $\rightarrow$ exit (take exit at $z = 0$, surface at $z = H$):

$$p_{\text{atm}} + 0 + \rho gH = p_{\text{atm}} + \frac{1}{2}\rho u_e^2 + 0.$$

Cancel p_{atm} and divide by ρ :

$$gH = \frac{1}{2}u_e^2.$$

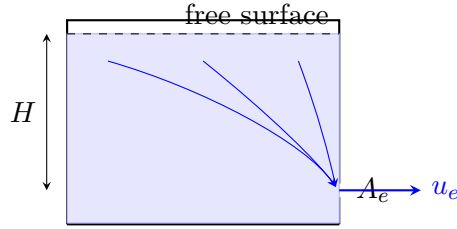
Solve for u_e :

$$u_e = \sqrt{2gH}.$$

Volume flux from continuity at the exit:

$$Q = A_e u_e = A_e \sqrt{2gH}.$$

$$\boxed{u_e = \sqrt{2gH}, \quad Q = A_e \sqrt{2gH}.}$$



Question 3 — Bifurcations and 2-D phase portrait

Problem. Analyse $\dot{x} = rx - 2x^2 + x^3$: fixed points, stability, bifurcations r_1, r_2 , normal forms. Then phase-portrait $\dot{x} = x^2 - y - 1$, $\dot{y} = (x - 2)y$.

Fixed point $x^* = 0$

Set $\dot{x} = 0$:

$$x(r - 2x + x^2) = 0.$$

$x = 0$ is always a root. Stability from $f'(x) = r - 4x + 3x^2$:

$$f'(0) = r.$$

Stable for $r < 0$, unstable for $r > 0$. Bifurcation at:

$$\boxed{r_1 = 0.}$$

Other fixed points

Other roots solve $x^2 - 2x + r = 0$:

$$x_{\pm} = \frac{2 \pm \sqrt{4 - 4r}}{2} = 1 \pm \sqrt{1 - r}.$$

Real for $r \leq 1$. Use $x_{\pm}^2 = 2x_{\pm} - r$ in $f'(x) = r - 4x + 3x^2$:

$$f'(x_{\pm}) = r - 4x_{\pm} + 3(2x_{\pm} - r).$$

Expand:

$$f'(x_{\pm}) = r - 4x_{\pm} + 6x_{\pm} - 3r.$$

Combine like terms:

$$f'(x_{\pm}) = -2r + 2x_{\pm} = 2(x_{\pm} - r).$$

Substitute $x_{\pm} = 1 \pm \sqrt{1-r}$:

$$f'(x_{\pm}) = 2(1 \pm \sqrt{1-r} - r) = 2((1-r) \pm \sqrt{1-r}).$$

Factor $\sqrt{1-r}$:

$$f'(x_{\pm}) = 2\sqrt{1-r}(\sqrt{1-r} \pm 1).$$

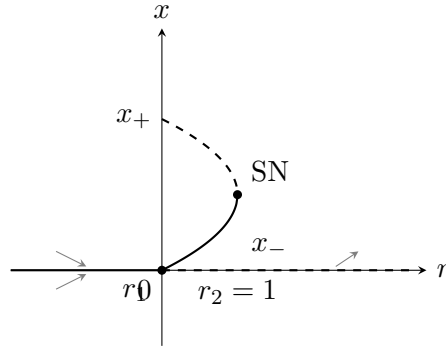
For $0 < r < 1$: $\sqrt{1-r} > 0$ and $\sqrt{1-r} - 1 < 0$, so $f'(x_-) < 0$ (stable); $\sqrt{1-r} + 1 > 0$, so $f'(x_+) > 0$ (unstable). Both vanish at $r = 1$.

Second bifurcation at:

$$r_2 = 1.$$

At $r = 1$: $x_{\pm} = 1$ collide (saddle-node). At $r = 0$: $x_- = 0$ collides with $x^* = 0$ (transcritical; $x_+ = 2$ persists for all $r \leq 1$).

Bifurcation diagram and normal forms



Solid = stable, dashed = unstable.

Normal form at $r_1 = 0$: drop the cubic x^3 (higher-order near $x = 0$):

$$\dot{x} \approx rx - 2x^2.$$

Fixed points: $x = 0$ and $x = r/2$ cross at $r = 0$ and exchange stability:

$$\dot{x} = rx - 2x^2 \quad (\text{transcritical}).$$

Normal form at $r_2 = 1$: let $r = 1 + \mu$, $x = 1 + y$ so $\dot{x} = \dot{y}$:

$$\dot{y} = (1 + \mu)(1 + y) - 2(1 + y)^2 + (1 + y)^3.$$

Expand $(1 + \mu)(1 + y)$:

$$(1 + \mu)(1 + y) = 1 + y + \mu + \mu y.$$

Expand $-2(1 + y)^2$:

$$-2(1 + y)^2 = -2 - 4y - 2y^2.$$

Expand $(1 + y)^3$:

$$(1 + y)^3 = 1 + 3y + 3y^2 + y^3.$$

Sum the three contributions:

$$\dot{y} = (1 - 2 + 1) + (1 - 4 + 3)y + \mu + \mu y + (-2 + 3)y^2 + y^3.$$

Collect (constants and linear in y both vanish):

$$\dot{y} = \mu + \mu y + y^2 + y^3.$$

Drop higher orders μy , y^3 near $(\mu, y) = (0, 0)$:

$$\dot{y} = \mu + y^2 \quad (\text{saddle-node}).$$

2-D phase portrait

Set $\dot{y} = 0$:

$$(x - 2)y = 0 \Rightarrow y = 0 \text{ or } x = 2.$$

On branch $y = 0$, set $\dot{x} = 0$:

$$x^2 - 1 = 0 \Rightarrow x = \pm 1.$$

On branch $x = 2$, set $\dot{x} = 0$:

$$4 - y - 1 = 0 \Rightarrow y = 3.$$

Fixed points: $(1, 0)$, $(-1, 0)$, $(2, 3)$.

General Jacobian:

$$J = \begin{pmatrix} \partial f_1 / \partial x & \partial f_1 / \partial y \\ \partial f_2 / \partial x & \partial f_2 / \partial y \end{pmatrix} = \begin{pmatrix} 2x & -1 \\ y & x - 2 \end{pmatrix}.$$

At $(1, 0)$:

$$J = \begin{pmatrix} 2 & -1 \\ 0 & -1 \end{pmatrix}, \quad \tau = 1, \Delta = -2.$$

$\Delta < 0$ implies saddle. Solve $\det(J - \lambda I) = (2 - \lambda)(-1 - \lambda) = 0$:

$$\lambda_1 = 2, \quad \lambda_2 = -1.$$

Eigenvector for $\lambda_1 = 2$: $(J - 2I)\mathbf{v} = 0$ gives $-v_2 = 0$, so $v = (1, 0)^T$ (unstable). Eigenvector for $\lambda_2 = -1$: $3v_1 - v_2 = 0$, so $v = (1, 3)^T$ (stable).

At $(-1, 0)$:

$$J = \begin{pmatrix} -2 & -1 \\ 0 & -3 \end{pmatrix}, \quad \tau = -5, \Delta = 6.$$

Compute discriminant:

$$\tau^2 - 4\Delta = 25 - 24 = 1 > 0.$$

With $\tau < 0$, $\Delta > 0$, $\tau^2 > 4\Delta$: stable node. Solve $\det(J - \lambda I) = (-2 - \lambda)(-3 - \lambda) = 0$:

$$\lambda_1 = -2, \quad \lambda_2 = -3.$$

Eigenvectors: $(J + 2I)\mathbf{v} = 0$ gives $v = (1, 0)^T$; $(J + 3I)\mathbf{v} = 0$ gives $v_1 - v_2 = 0$, so $v = (1, 1)^T$.

At $(2, 3)$:

$$J = \begin{pmatrix} 4 & -1 \\ 3 & 0 \end{pmatrix}, \quad \tau = 4, \Delta = 3.$$

Compute discriminant:

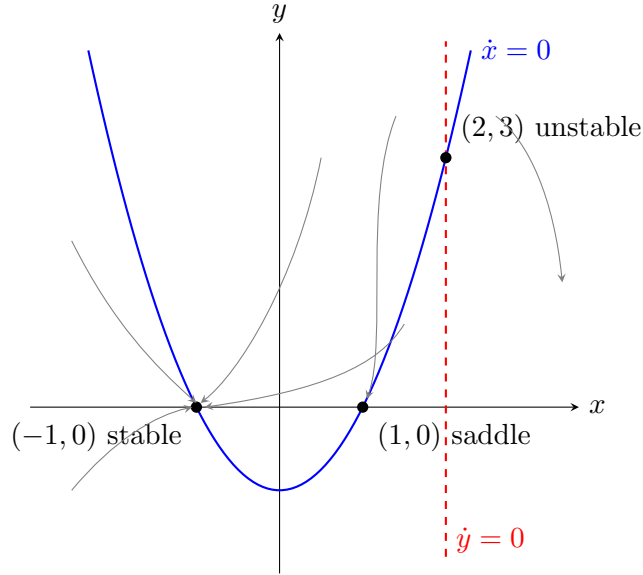
$$\tau^2 - 4\Delta = 16 - 12 = 4 > 0.$$

With $\tau > 0$, $\Delta > 0$, $\tau^2 > 4\Delta$: unstable node. Solve $\lambda^2 - 4\lambda + 3 = 0$:

$$\lambda_1 = 1, \quad \lambda_2 = 3.$$

Eigenvector for $\lambda_1 = 1$: $3v_1 - v_2 = 0$, so $v = (1, 3)^T$. Eigenvector for $\lambda_2 = 3$: $v_1 - v_2 = 0$, so $v = (1, 1)^T$.

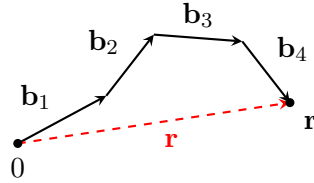
Nullclines: $\dot{x} = 0$ on parabola $y = x^2 - 1$; $\dot{y} = 0$ on $y = 0$ and $x = 2$.



Question 4 — FJC, WLC and DNA

Problem. Show $\langle r^2 \rangle = Nb^2$ for FJC; derive WLC formula and limits; obtain Gaussian end-to-end distribution; show entropic spring $A = a + cr^2$ with c in terms of N, b, T ; high-force regime; numerical estimates for ds and ssDNA.

FJC: $\langle r^2 \rangle = Nb^2$



End-to-end vector is the sum of N segments of length b :

$$\mathbf{r} = \sum_{i=1}^N \mathbf{b}_i, \quad |\mathbf{b}_i| = b.$$

Square and split diagonal/off-diagonal:

$$\mathbf{r} \cdot \mathbf{r} = \sum_{i=1}^N \mathbf{b}_i \cdot \mathbf{b}_i + \sum_{i \neq j} \mathbf{b}_i \cdot \mathbf{b}_j.$$

Evaluate diagonal terms ($\mathbf{b}_i \cdot \mathbf{b}_i = b^2$):

$$\mathbf{r} \cdot \mathbf{r} = Nb^2 + \sum_{i \neq j} \mathbf{b}_i \cdot \mathbf{b}_j.$$

Take ensemble average; non-interacting segments give $\langle \mathbf{b}_i \cdot \mathbf{b}_j \rangle = 0$ for $i \neq j$:

$\langle r^2 \rangle = Nb^2.$

WLC: derivation and limits

Continuous tangent:

$$\mathbf{r} = \int_0^L \mathbf{t}(s) ds, \quad \langle \mathbf{t}(s) \cdot \mathbf{t}(s') \rangle = e^{-|s-s'|/P}.$$

Square and average:

$$\langle r^2 \rangle = \int_0^L \int_0^L e^{-|s-s'|/P} ds ds'.$$

Use symmetry $s' < s$, factor of 2:

$$\langle r^2 \rangle = 2 \int_0^L ds \int_0^s e^{-(s-s')/P} ds'.$$

Substitute $u = s - s'$ in the inner integral:

$$\int_0^s e^{-(s-s')/P} ds' = \int_0^s e^{-u/P} du = P(1 - e^{-s/P}).$$

Insert into outer integral:

$$\langle r^2 \rangle = 2P \int_0^L (1 - e^{-s/P}) ds.$$

Evaluate the outer integral:

$$\int_0^L (1 - e^{-s/P}) ds = L + P(e^{-L/P} - 1).$$

Hence:

$$\langle r^2 \rangle = 2P[L + P(e^{-L/P} - 1)].$$

$$\langle r^2 \rangle = 2P^2 \left(e^{-L/P} - 1 + \frac{L}{P} \right).$$

Limit $L \gg P$: $e^{-L/P} \rightarrow 0$:

$$\langle r^2 \rangle \rightarrow 2PL.$$

Identifies effective Kuhn length $b_K = 2P$ and segment count $L/(2P)$: Gaussian-coil regime.

Limit $L \ll P$: expand $e^{-L/P} = 1 - L/P + L^2/(2P^2) - \dots$:

$$e^{-L/P} - 1 + L/P \approx L^2/(2P^2).$$

Multiply by $2P^2$:

$$\langle r^2 \rangle \rightarrow L^2.$$

Rigid-rod regime; chain too short to bend.

End-to-end distribution and width

3-D Gaussian random walk: x, y, z each diffuse independently in N steps. For an isotropic random step, $\langle b_x^2 \rangle = b^2/3$, so identify $2Dt \rightarrow Nb^2/3$ per axis:

$$P_1(x_i, N) = \sqrt{\frac{3}{2\pi Nb^2}} \exp\left(-\frac{3x_i^2}{2Nb^2}\right).$$

Multiply three independent Gaussians $P(\mathbf{r}, N) = P_1(x)P_1(y)P_1(z)$:

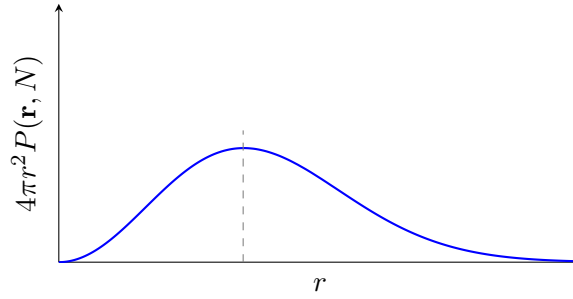
$$P(\mathbf{r}, N) = \left(\frac{3}{2\pi N b^2} \right)^{3/2} \exp\left(-\frac{3(x^2 + y^2 + z^2)}{2N b^2} \right).$$

Use $r^2 = x^2 + y^2 + z^2$:

$$P(\mathbf{r}, N) = \left(\frac{3}{2\pi N b^2} \right)^{3/2} \exp\left(-\frac{3r^2}{2N b^2} \right).$$

$$\boxed{P(\mathbf{r}, N) = \left(\frac{3}{2\pi N b^2} \right)^{3/2} \exp\left(-\frac{3r^2}{2N b^2} \right).}$$

Width along one axis $\sigma = \sqrt{N b^2 / 3}$. Total $\langle r^2 \rangle = 3\sigma^2 = N b^2$, recovering the FJC result.



Entropic spring

Boltzmann entropy of conformations with end-to-end vector $\mathbf{r}$:

$$S(\mathbf{r}) = k_B \ln \Omega(\mathbf{r}) = k_B \ln [V P(\mathbf{r}, N)].$$

Insert Gaussian and split into r -independent / r -dependent parts:

$$S(\mathbf{r}) = k_B \ln C - \frac{3k_B r^2}{2N b^2},$$

with C collecting r -independent prefactors. At fixed T , with internal energy U independent of $\mathbf{r}$:

$$A(\mathbf{r}) = U - T S(\mathbf{r}).$$

Substitute $S(\mathbf{r})$:

$$A(\mathbf{r}) = U - T k_B \ln C + \frac{3k_B T}{2N b^2} r^2.$$

Identify $a = U - T k_B \ln C$, $c = 3k_B T / (2N b^2)$:

$$\boxed{A(\mathbf{r}) = a + c r^2, \quad c = \frac{3k_B T}{2N b^2}.}$$

Quadratic in $r \Rightarrow$ Hookean restoring force $\mathbf{F} = -\nabla A = -2c\mathbf{r}$. Stiffness comes purely from entropy (more conformations available at small $|\mathbf{r}|$): an *entropic spring*.

High-force limit

Each segment couples to applied force $\mathbf{F} = F\hat{\mathbf{z}}$; one-segment partition function with $d\Omega = \sin\theta d\theta d\varphi$:

$$Z_1 = \int_0^{2\pi} d\varphi \int_0^\pi \frac{\sin\theta d\theta}{4\pi} e^{\beta F b \cos\theta}.$$

Evaluate (let $u = \cos\theta$):

$$Z_1 = \frac{1}{2} \int_{-1}^1 e^{\beta F b u} du = \frac{\sinh(\beta F b)}{\beta F b}.$$

Mean projection per segment via $\langle b_z \rangle = \partial \ln Z_1 / \partial(\beta F)$:

$$\langle b_z \rangle = b \mathcal{L}(\beta F b), \quad \mathcal{L}(x) = \coth x - \frac{1}{x}.$$

Total extension over N segments:

$$z = N b \mathcal{L}(\beta F b).$$

High-force limit $\beta F b \gg 1$: $\coth(\beta F b) \rightarrow 1$:

$$\mathcal{L}(\beta F b) \rightarrow 1 - \frac{1}{\beta F b} = 1 - \frac{k_B T}{F b}.$$

Hence:

$$z \rightarrow N b \left(1 - \frac{k_B T}{F b} \right).$$

Solve for F :

$$1 - \frac{z}{N b} = \frac{k_B T}{F b}.$$

Rearrange:

$$F \simeq \frac{k_B T}{b} \frac{1}{1 - z/(N b)}.$$

F diverges as the chain approaches contour length $L = N b$.

Bacterial DNA — numerical estimates

Double-stranded: contour length

$$L = 5 \times 10^6 \times 0.34 \text{ nm} = 1.7 \times 10^6 \text{ nm} = 1.7 \text{ mm}.$$

Persistence length:

$$P = 150 \times 0.34 \text{ nm} = 51 \text{ nm}.$$

Use $L \gg P$ limit $\langle r^2 \rangle \approx 2 P L$:

$$\langle r^2 \rangle \approx 2(51)(1.7 \times 10^6) \text{ nm}^2 = 1.73 \times 10^8 \text{ nm}^2.$$

Take square root:

$$\sqrt{\langle r^2 \rangle}_{\text{ds}} \approx 1.32 \times 10^4 \text{ nm} = 13.2 \text{ } \mu\text{m}.$$

Single-stranded: persistence length

$$P_{\text{ss}} = 5 \times 0.34 \text{ nm} = 1.7 \text{ nm}.$$

Apply $\langle r^2 \rangle \approx 2P_{ss}L$:

$$\langle r^2 \rangle \approx 2(1.7)(1.7 \times 10^6) \text{ nm}^2 = 5.78 \times 10^6 \text{ nm}^2.$$

Take square root:

$$\sqrt{\langle r^2 \rangle}_{ss} \approx 2.40 \times 10^3 \text{ nm} = 2.4 \mu\text{m}.$$

Probability ratio $P(R=0)/P(R=50 \mu\text{m})$. Use Gaussian with $Nb^2 \rightarrow \langle r^2 \rangle$:

$$\frac{P(0)}{P(R)} = \exp\left(\frac{3R^2}{2\langle r^2 \rangle}\right).$$

ds case: $R = 50 \mu\text{m} = 5 \times 10^4 \text{ nm}$, so $R^2 = 2.5 \times 10^9 \text{ nm}^2$. Substitute $\langle r^2 \rangle = 1.73 \times 10^8 \text{ nm}^2$:

$$\frac{3R^2}{2\langle r^2 \rangle} = \frac{3(2.5 \times 10^9)}{2(1.73 \times 10^8)}.$$

Evaluate:

$$\frac{3R^2}{2\langle r^2 \rangle} \approx 21.7.$$

Exponentiate:

$$P(0)/P(50 \mu\text{m})_{ds} \approx e^{21.7} \approx 2.7 \times 10^9.$$

ss case: substitute $\langle r^2 \rangle = 5.78 \times 10^6 \text{ nm}^2$:

$$\frac{3R^2}{2\langle r^2 \rangle} = \frac{3(2.5 \times 10^9)}{2(5.78 \times 10^6)}.$$

Evaluate:

$$\frac{3R^2}{2\langle r^2 \rangle} \approx 649.$$

Exponentiate:

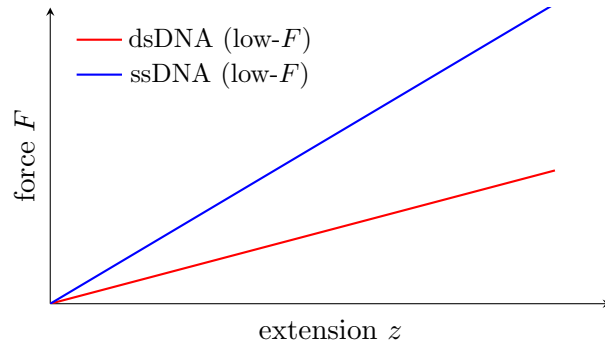
$$P(0)/P(50 \mu\text{m})_{ss} \approx e^{649} \sim 10^{282}.$$

Comment: ssDNA is far more compact than dsDNA (smaller P , smaller $\langle r^2 \rangle$); reaching $50 \mu\text{m}$ is astronomically improbable. Both ratios $\gg 1$: extended states are heavily entropically suppressed.

Low-force regime. Both behave as Hookean entropic springs with stiffness $k = 3k_B T / \langle r^2 \rangle$:

$$F = \frac{3k_B T}{\langle r^2 \rangle} z.$$

ssDNA has smaller $\langle r^2 \rangle \Rightarrow$ steeper slope (stiffer at low force). dsDNA has larger $\langle r^2 \rangle \Rightarrow$ softer at low force.



ssDNA (steeper) is entropically stiffer because its smaller persistence length yields a more compact equilibrium coil and thus a larger $3k_B T / \langle r^2 \rangle$.